

## Customer Churn Case Study with AI

A step-by-step guide covering slides 18-25: load and describe the Telco churn data in Colab, research the problem in NotebookLM, test hypotheses with AI-generated charts, and build an interactive Gradio dashboard.

**Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.**

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### Before you start

You need a Google account (for Colab and NotebookLM). Download Telco-Customer-Churn.csv from Canvas first. Everything runs in your browser - no installation required. If you need a refresher on opening Colab and using the Gemini agent, see the Week 2 walkthroughs.

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| <b>Slide 18</b>     | Upload the CSV to Colab; AI reads and describes the data |
| <b>Slides 19-20</b> | Research churn in NotebookLM and read the AI mind map    |
| <b>Slides 21-22</b> | Discuss and form hypotheses about what drives churn      |
| <b>Slide 23</b>     | AI generates charts to test each hypothesis              |
| <b>Slide 25</b>     | Build an interactive Gradio dashboard                    |

*Two companion screen recordings (Week3\_Churn\_Part1\_Colab\_NotebookLM.gif and Week3\_Churn\_Part2\_Hypotheses\_Gradio.gif) show every click in sequence.*

### Step 1 - Load the data in Colab (Slide 18)

Open a new Colab notebook, open the Files panel (folder icon), and upload Telco-Customer-Churn.csv. Then open the Gemini panel and ask it to read the file:

```
help me read the Telco-Customer-Churn.csv to a pandas dataframe
```

Gemini writes the pandas code and shows the first rows. Click Accept & Run.

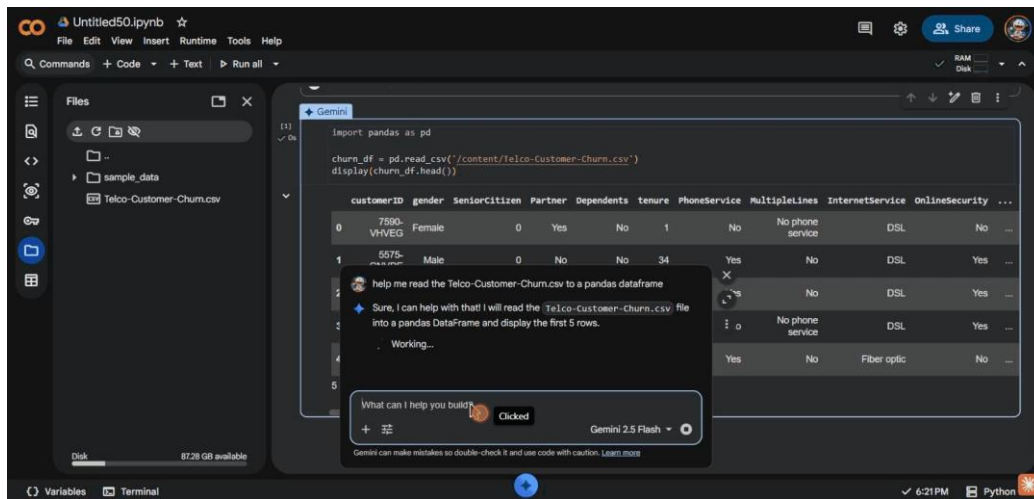


Figure 1 - The CSV in the Files panel; Gemini generates the code to read it into a DataFrame.

## Step 2 - Describe the columns (Slide 18)

Ask the agent to explain the data so you understand each field before analysing it:

give me a table explain the meaning of each column and its data type

The result lists all 21 columns - for example tenure (months with the company), Contract (Month-to-month / One year / Two year), and Churn, which is the target variable you want to predict.

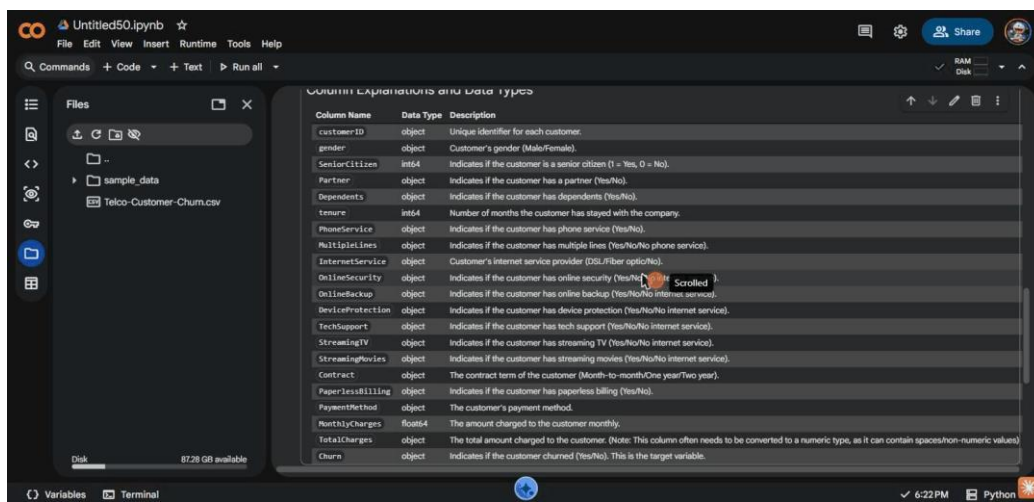


Figure 2 - The AI-generated column dictionary: name, data type and meaning for every field.

## Step 3 - Research the problem in NotebookLM (Slide 19)

Go to notebooklm.google.com and create a new notebook. In the source search box, choose Web + Fast Research and ask for industry background:

I am working on a data science project on telco customer churn.

Please provide industry insights and tell me what features I need to analyse to understand churn.

NotebookLM searches the web, returns a set of relevant sources, and you click Import to add them to the notebook.

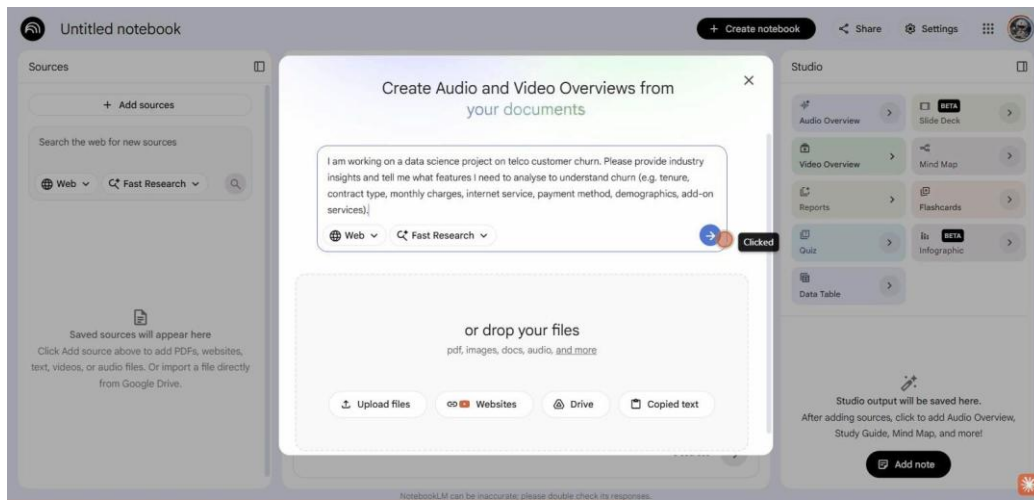


Figure 3 - Running Fast Research in NotebookLM to gather sources about telco churn.

## Step 4 - Read the AI mind map (Slide 20)

In the Studio panel, click Mind Map. NotebookLM builds a visual map from your sources. Expand Key Churn Drivers to see the main factors.

Use this map and discuss with your classmates: which factors do you believe most affect churn? Here the drivers are contract type, internet service, tenure, payment method and the monthly/total charges.

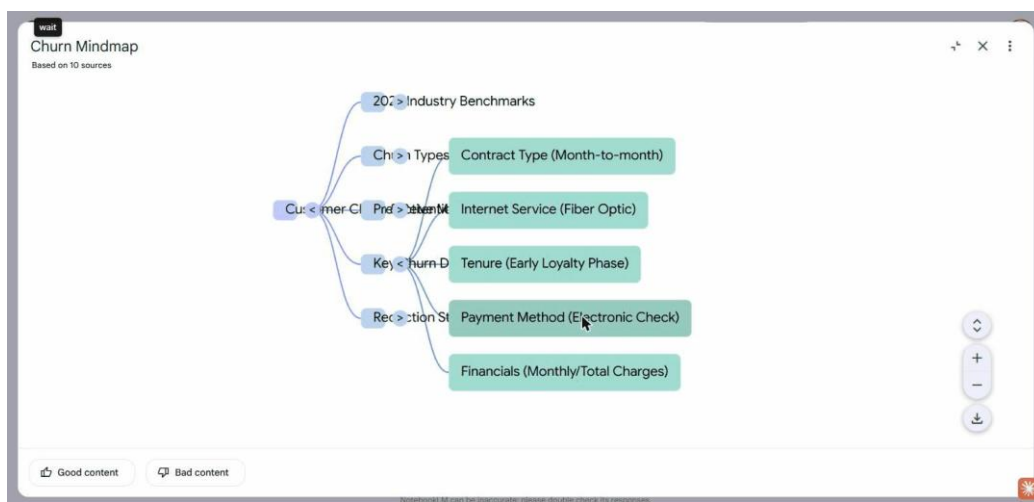


Figure 4 - The Churn Mindmap with the Key Churn Drivers branch expanded.

## Step 5 - Form and test hypotheses (Slides 22-23)

Turn the drivers into testable hypotheses, then back in Colab ask the agent to check them against your data:

generate the python code to help me examine the following hypotheses (if applicable to our dataset): 1. Customers with longer tenure have lower churn. 2. Month-to-month contracts have higher churn than longer contracts. 3. Fiber optic internet service has higher churn. 4. Customers without tech support or online security churn more.

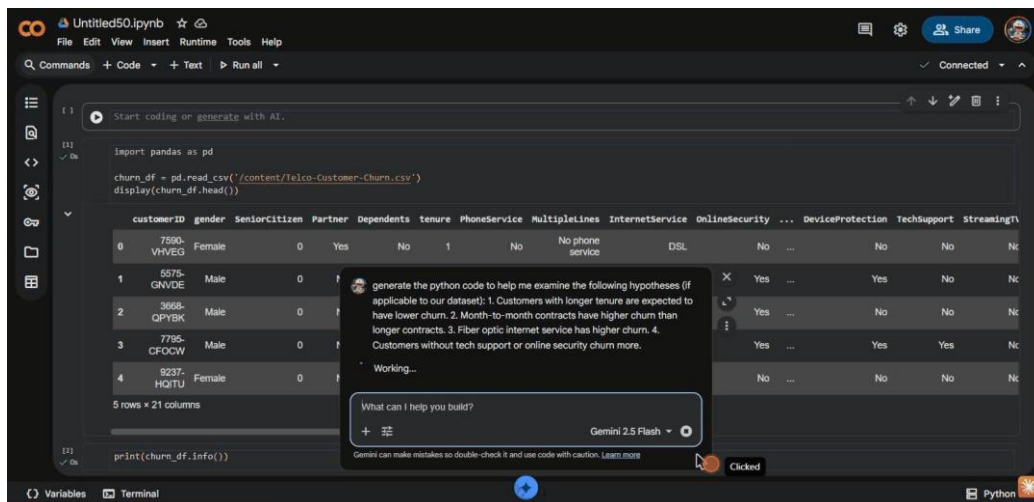


Figure 5 - Sending the hypotheses to the Gemini agent to generate the analysis code.

## Step 6 - Read the charts (Slide 23)

Click Accept & Run. The notebook produces a chart per hypothesis. The data strongly supports them - for example month-to-month contracts churn at about 43% versus under 3% for two-year contracts, and fiber-optic customers churn far more than DSL or no-internet customers.

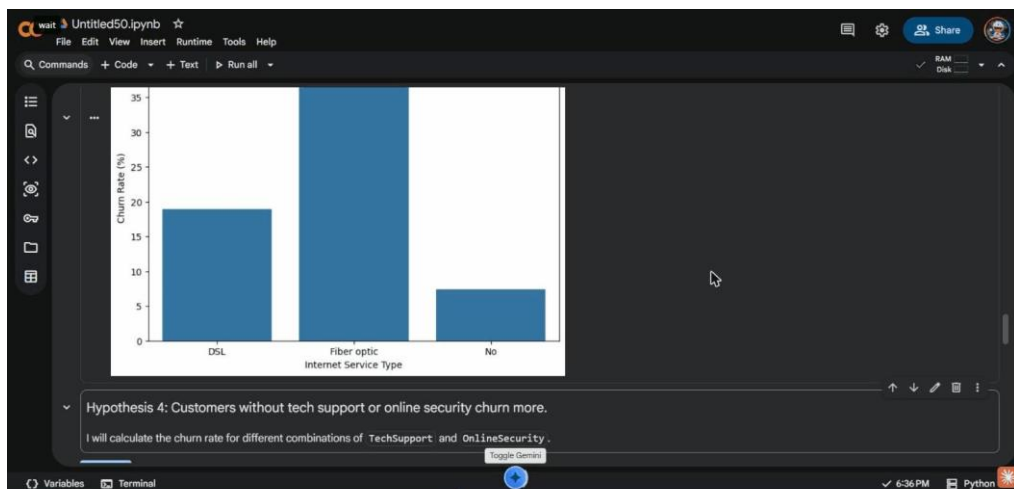


Figure 6 - Churn rate by internet service type - fiber optic is clearly the highest.

Reflection: do you trust the generated code and the results? Do they support your hypotheses? If not, what might be the issue?

## Step 7 - Build an interactive dashboard (Slide 25)

Finally, turn the analysis into an interactive app. Ask the agent:

Build a dashboard using Gradio to help me examine the visualisations and hypotheses above in an interactive way

After Accept & Run, Colab installs Gradio and launches the dashboard inline, with one tab per hypothesis so you can explore each chart interactively.

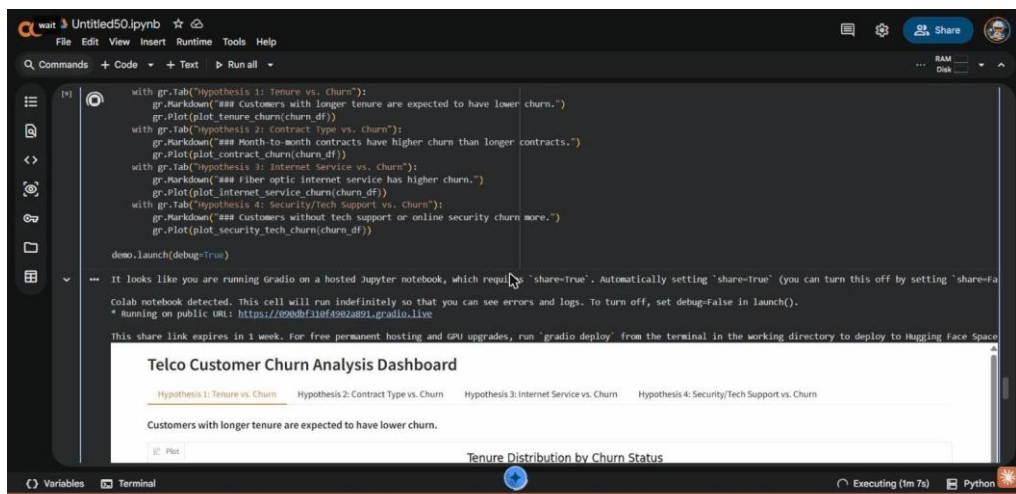


Figure 7 - The Gradio dashboard with a tab for each hypothesis, running inside Colab.

## Good to know

- Always review AI-generated code before trusting it. Colab warns: Gemini can make mistakes so double-check it and use code with caution.
- Use your file's exact name in prompts. If the AI cannot find the file, re-check the name in the Files panel.
- Uploaded files and the runtime are temporary. If the session drops, re-upload the CSV and re-run the cells; in NotebookLM your notebook and sources are saved automatically.
- The Gradio cell keeps running so the app stays live, and prints a temporary public share link that expires after a week.
- Each student can run their own NotebookLM session and compare the factors their mind map surfaces.